Feb3 Quiz

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① D: $P_d = 1700 - 20Q \rightarrow$ x-int $= \frac{-c}{m} = \frac{-1700}{-20} = 85$

S: $P_s = 500 + 40Q \rightarrow$ x-int $= \frac{-c}{m} = \frac{-500}{40} = -12.5$

(a) (P_e, q_e)?

At e:

$P_e = P_d = P_s$ & $q_e = q_s = q_d$

$\Downarrow$

$P_d = P_s$

$1700 - 20Q = 500 + 40Q$

$1700 - 500 = 40Q + 20Q$

$1200 = 60Q$

$Q_e = \frac{1200}{60} = 20$

$P_e = 500 + 40(20) = 1300$

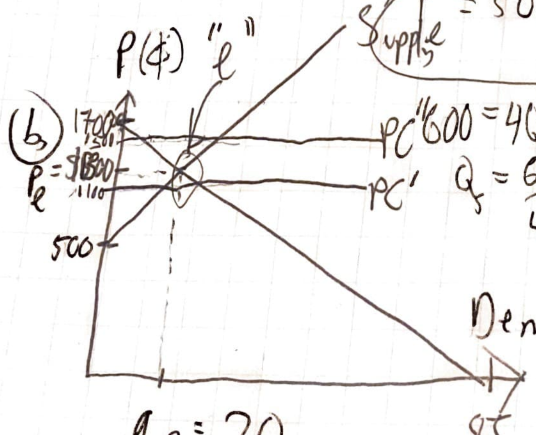
$20Q = 1700 - 1100$

$20Q = 600$

$Q_D = 30$

$(Q_s - Q_D) = -15$ barrels

(b)



PC'' $600 = 40Q$

PC' $Q_s = \frac{600}{40} = 15$

(c) Yes, $\because$ $PC' = 1100$ is less than $P_e = 1300$!

$\therefore$ Shortage of Milk of 15 barrels

S: $1100 = 500 + 40Q$

D: $1100 = 1700 - 20Q$

(d) No, $\because$ $PC'' = 1500 > P_e \rightarrow$ consumers won't pay at that high of a price $\rightarrow (P_e, q_e)$ is chosen by market instead